

CB-SAFE: Code-Based Post-Quantum Secure Aggregation for Byzantine-Robust Federated Learning

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Abstract—Federated learning (FL) keeps raw data on-device, yet the transmitted model updates leak private information, and secure aggregation, which masks individual updates so the server sees only their sum, has become the standard mitigation. This creates two open problems. First, the masks rest on classical key exchange that a quantum computer breaks, and the few post-quantum FL systems are almost exclusively lattice-based, concentrating systemic risk on a single hardness assumption. Second, hiding individual updates denies Byzantine-robustness defenses the visibility they need. We critically review 20 recent high-quality papers spanning secure aggregation, post-quantum FL, and Byzantine-robust FL, and show that no existing framework simultaneously provides code-based post-quantum confidentiality, Byzantine robustness, per-client identification, and scalability. To close this gap we propose CB-SAFE, a KEM-agnostic secure-aggregation protocol instantiated with the NIST-selected code-based KEM HQC, robust aggregation over cluster sums, and CB-SAFE+, a temporal group-testing defense that identifies hidden attackers from privacy-preserving observations. We detail the methodology and an evaluation plan over four datasets and three attack families.

Index Terms—Federated learning, secure aggregation, post-quantum cryptography, code-based KEM, Byzantine robustness, group testing.

I. INTRODUCTION

Federated learning (FL) trains a shared model across many clients without centralizing their data [1], but the transmitted updates are not private: gradient-inversion attacks reconstruct training samples from a single update [2]. Secure aggregation closes this channel by ensuring the server observes only the *sum* of updates, using pairwise masks that cancel on aggregation [3]. The confidentiality of those masks, however, rests on key establishment that Shor’s algorithm breaks once a cryptographically relevant quantum computer exists [29], and traffic recorded today can be decrypted later. The standard fix replaces classical key exchange with a post-quantum KEM; nearly all post-quantum FL does so with *lattice* cryptography, concentrating an entire ecosystem’s confidentiality on one assumption. In March 2025 NIST selected the *code-based* KEM HQC as a diversified backup to the lattice standard [30]; that diversity has not yet reached FL.

A second, orthogonal tension is that secure aggregation and Byzantine robustness pull in opposite directions: hiding individual updates removes exactly the per-client signal that

poisoning defenses rely on. This proposal update (i) synthesizes the recent literature across these three areas, (ii) establishes the resulting research gap, and (iii) develops a detailed methodology for CB-SAFE, the first code-based post-quantum secure-aggregation framework that is also Byzantine-robust, identifies malicious clients, and scales beyond a fixed small population.

II. LITERATURE REVIEW

The literature relevant to CB-SAFE falls into four themes: secure aggregation, post-quantum federated learning, Byzantine-robust aggregation, and their intersection. We review the most pertinent recent work in each and, rather than summarizing studies individually, compare them critically by what each approach does, how it works, and where it falls short. Table I consolidates this comparison.

A. Secure Aggregation

Bonawitz *et al.* [3] introduced masking-based secure aggregation with dropout tolerance via secret-shared seeds; Bell *et al.* [4] reduced its quadratic cost to (poly)logarithmic, and LightSecAgg [5] and SwiftAgg [6] further lower communication with coded one-shot recovery. Flamingo [7] removes the per-round setup for the multi-round FL setting. *Critique*: this line steadily improves efficiency, but every protocol reveals only the aggregate, which can itself leak information about participants [8], and, crucially, hides the per-client updates that robustness defenses require. None is post-quantum.

B. Post-Quantum Federated Learning

PQSF [9] reconstructs masks with lattice secret sharing; Qin and Xu [10] build single-round aggregation from a lattice key-homomorphic PRF; a 2026 framework [11] pairs reputation-weighted robustness with lattice-based aggregation; and CQSA [12] performs clustered aggregation on near-term *quantum hardware*. *Critique*: confidentiality is uniformly lattice- or quantum-hardware-based; a single lattice cryptanalytic advance would break the whole class, and most such systems either omit Byzantine robustness or assume hardware that is not yet deployable. No code-based instantiation exists.

C. Byzantine-Robust Aggregation

Robust rules replace the mean: Multi-Krum [13], coordinate-wise trimmed mean and median [14], Bulyan [15], and the geometric median (RFA) [16]. Trust- and detection-based methods add FLTrust [17], FLDetector [18], and the backdoor defense FLAME [19], motivated by strong poisoning attacks [20]. *Critique*: all of these operate on *individual*, cleartext updates, the precise information secure aggregation removes, so they are fundamentally incompatible with a privacy-preserving pipeline, and coordinate-wise statistics are provably fragile against amplified poisoning.

D. Robustness Under Secure Aggregation

The closest works combine both goals. RoFL [21] enforces zero-knowledge norm/well-formedness proofs on masked updates; EIFFeL [22] verifies secret-shared integrity checks; ELSA [23] uses two-server distributed trust robust to malicious clients; BREA [24] secret-shares distance computations; and PriRoAgg [25] adds privacy to robust aggregation. FedGT [26] is nearest to our mechanism: it identifies malicious clients by non-adaptive group testing [27], [28] over overlapping secure-aggregation groups. *Critique*: all are classical (pre-quantum); RoFL/EIFFeL/ELSA *bound or validate* updates rather than identifying attackers; and FedGT's parity-check code is fixed for a small population ($n \leq 31$) and over-excludes honest clients under amplified attacks.

Synthesis. Across the literature, confidentiality and robustness have been advanced largely in isolation: secure-aggregation and post-quantum work protect privacy but suppress the signal defenses need, while robust-aggregation work assumes cleartext updates. The handful of systems that reconcile the two remain classical and either bound updates instead of identifying attackers (RoFL, EIFFeL, ELSA) or, like FedGT, identify attackers only within a fixed small population, as the limitation column of Table I makes clear.

III. RESEARCH GAP AND PROBLEM STATEMENT

Gap. No FL framework simultaneously provides (1) *code-based* post-quantum confidentiality (a hedge against lattice-only monoculture), (2) secure aggregation, (3) Byzantine robustness, (4) explicit per-client identification of attackers, and (5) scalability beyond a fixed small client count. Table II makes this explicit as a capability matrix: every prior family misses at least one axis, and only CB-SAFE+ satisfies all six.

Problem statement. Given N clients of which a fraction f are Byzantine, train a global model such that (a) the server learns only aggregate information under a post-quantum, code-based confidentiality assumption, and (b) malicious clients are identified and excluded despite never being observed individually. A key obstacle we target is a *laundering* effect: averaging poisoned and honest updates within a privacy cluster turns an amplified sign-flip attack into a plausible-magnitude, direction-inverted cluster sum that defeats coordinate-wise robust rules, so robustness must be recovered *without* breaking the privacy that causes it.

IV. PROPOSED METHODOLOGY

A. Overview and Per-Round Workflow

CB-SAFE augments standard federated averaging with three protective layers (a code-based post-quantum masking layer, a clustering layer that trades privacy against robustness in a controlled way, and a temporal detection layer) while keeping the server's view limited to cluster-level sums. The end-to-end pipeline is shown in Fig. 1. Before training, each client performs a *one-time* HQC key encapsulation to establish long-lived pairwise secrets. Then, in every round r : (1) each client trains the global model locally for one epoch and blinds its update with pairwise masks; (2) clients are partitioned into k clusters of size c and secure aggregation runs *within* each cluster, so the server receives only k cluster sums, never an individual update; (3) the server applies a robust aggregation rule *across* those cluster sums to update the global model; and (4) a loss/direction probe on each cluster sum feeds a temporal group-testing procedure that accumulates per-client suspicion and, after a warmup, excludes identified attackers. Steps 1–4 repeat with a *freshly randomized* clustering every round, which is exactly what makes attacker identification possible without de-anonymizing any single client. Algorithm 1 summarizes the server-side round.

B. Code-Based Post-Quantum Masked Aggregation

The confidentiality layer generalizes Bonawitz-style double masking [3] to run over *any* IND-CCA2 KEM through a KEM-agnostic interface. Each pair of clients derives a shared secret once, via HQC key encapsulation; per-round mask seeds are expanded from that secret with HKDF-SHA256 and realized as ChaCha20 keystreams that cancel exactly when cluster members' updates are summed, so the server recovers the cluster sum but learns nothing about any single update. Because encapsulation occurs only at setup, the code-based premium (HQC's larger keys/ciphertexts and slower operations relative to lattice KEMs) is a *one-time* cost amortized over all training rounds, while per-round communication is KEM-independent. Substituting ML-KEM for HQC changes exactly one module, letting us report both at matched NIST levels 1/3/5 and confirm that the two produce numerically identical models.

C. Clustering and the Privacy–Robustness Trade-off

Secure aggregation alone hides individual updates but blinds robust aggregation: a single global sum cannot be filtered for outliers. CB-SAFE resolves this by aggregating securely *within* clusters of size c and applying a robust rule (trimmed mean, median, or Multi-Krum) *across* the k resulting cluster sums. This exposes a tunable dial: each client's anonymity set is c , while the probability that a cluster contains none of the fN attackers decays as $(1 - f)^c$. Larger c strengthens privacy but raises the chance a cluster is contaminated. Critically, we identify a *laundering* effect: an amplified sign-flip attacker inside a cluster yields a plausible-magnitude, direction-inverted cluster sum that coordinate-wise robust rules cannot separate from an honest one. Recovering robustness therefore requires

TABLE I
CRITICAL REVIEW OF REPRESENTATIVE PRIOR WORK ACROSS THE THREE THEMES: WHAT EACH LINE OF WORK DOES AND HOW, AND ITS KEY LIMITATION RELATIVE TO THIS PROJECT.

Study / category	Approach: what they did and how	Key limitation(s)
Secure aggregation [3], [4], [7]	Mask each client update with pairwise secrets that cancel on summation, so the server learns only the aggregate; dropout tolerance via secret-shared seeds, with (poly)logarithmic-overhead and single-server multi-round variants.	Classical key exchange (not post-quantum); by hiding per-client updates it removes exactly the signal that robustness defenses need.
Coordinate-wise robust rules [13]–[16]	Replace the mean with trimmed mean, median, Multi-Krum, or the geometric median to bound the influence of outlier updates on the aggregate.	Require cleartext per-client updates (incompatible with secure aggregation); provably fragile to amplified/laundered sign-flip poisoning.
Malicious-client detection [17]–[19]	Flag or down-weight suspicious clients using a trusted root, update-consistency over rounds, or clustering of gradients.	Inspect individual updates, breaking privacy; classical only; heuristic thresholds an adaptive attacker can evade.
Robustness under secure aggregation [21]–[23]	Enforce constraints on <i>masked</i> updates via zero-knowledge norm/well-formedness proofs, verifiable secret sharing, or two-server distributed trust.	Classical (pre-quantum); <i>bound or validate</i> updates rather than identifying which clients are malicious.
Group-testing identification (FedGT) [26]	Identify malicious clients by non-adaptive group testing over overlapping secure-aggregation groups, decoded with a per-round BCJR decoder.	Classical; parity-check code fixed for $n \leq 31$ clients; over-excludes honest clients under laundering.
Post-quantum FL [9]–[12]	Replace classical key exchange with lattice KEMs/secret sharing (or quantum hardware) for post-quantum confidential aggregation.	Lattice/quantum-only (monoculture risk); most omit Byzantine robustness; no code-based instantiation.
CB-SAFE+ (this work)	Code-based (HQC) KEM-agnostic masked cluster aggregation, robust aggregation across cluster sums, and temporal group testing to identify hidden attackers.	Unifies code-based post-quantum confidentiality, robustness, per-client identification, and scaling in a single framework.

TABLE II
CAPABILITY MATRIX ACROSS SIX AXES. $\checkmark$ = SUPPORTED, $\times$ = NOT, $\sim$ = PARTIAL.

Approach	Post-quantum	Code-based	Secure agg.	Byz.-robust	Identifies	Scales $N > 31$
Secure agg. [3], [4]	$\times$	$\times$	$\checkmark$	$\times$	$\times$	$\checkmark$
Robust rules [13], [14]	$\times$	$\times$	$\times$	$\checkmark$	$\times$	$\checkmark$
FLDetector [18]	$\times$	$\times$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$
RoFL/EIFFel/ELSA [21]–[23]	$\times$	$\times$	$\checkmark$	$\checkmark$	$\times$	$\checkmark$
FedGT [26]	$\times$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$	$\times$
Lattice PQ-FL [9], [11]	$\checkmark$	$\times$	$\checkmark$	$\sim$	$\times$	$\checkmark$
CB-SAFE+ (ours)	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$

identifying the responsible clients, not merely trimming outliers.

D. CB-SAFE+: Temporal Group-Testing Detection

CB-SAFE+ turns the per-round cluster sums into a non-adaptive group test spread over time. Against a small server-held root dataset $\mathcal{D}_0$ (200 samples, disjoint from all clients), each cluster sum yields a binary flag indicating whether it drives the model in a harmful direction (rising root loss or inverted gradient). Because the clustering is re-randomized each round, a fixed attacker is repeatedly placed in different, overlapping groups; combinatorial (COMP) decoding of the accumulated flags concentrates suspicion on the clients that are consistently “defective.” After a warmup of W rounds, clients whose suspicion exceeds a threshold τ (or an adaptive gap rule) are excluded for the remainder of training. Because the flags are post-processing of *already-revealed* cluster sums, CB-SAFE+ adds no update-content leakage beyond CB-

Algorithm 1 CB-SAFE round r (server view)

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1: Input: global model  $\theta$ ; clusters  $\{C_j\}_{j=1}^k$  of size  $c$ ; root set  $\mathcal{D}_0$ ; threshold  $\tau$ ; warmup  $W$ 
2: for each client  $i$  (in parallel) do
3:    $\delta_i \leftarrow \text{LOCALTRAIN}(\theta)$ ; send masked  $\delta_i$  within its cluster
4: end for
5: for each cluster  $j$  do
6:   server obtains cluster sum  $S_j = \sum_{i \in C_j} \delta_i$   $\triangleright$  masks cancel
7:    $g_j \leftarrow \text{loss/direction flag of } S_j \text{ against } \mathcal{D}_0$ 
8: end for
9:  $\theta \leftarrow \theta + \text{ROBUSTAGG}(\{S_j/c\})$   $\triangleright$  trimmed mean / median / Multi-Krum
10: if  $r \geq W$  then
11:   update per-client suspicion by COMP decoding of flags  $\{g_j\}$  over the re-randomized group assignment  $\triangleright$  temporal group testing
12:   exclude clients with suspicion  $> \tau$ 
13: end if
14: re-randomize clustering for round  $r+1$ 
15: Output: updated  $\theta$ , exclusion set

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SAFE: attackers are identified without ever being observed individually.

E. Datasets, Baselines, and Protocol

Datasets. CIFAR-10, FashionMNIST, EMNIST (image), and Edge-IIoTset (tabular IoT intrusion detection), partitioned across $N=30$ clients by a Dirichlet($\alpha=0.5$) non-IID split.

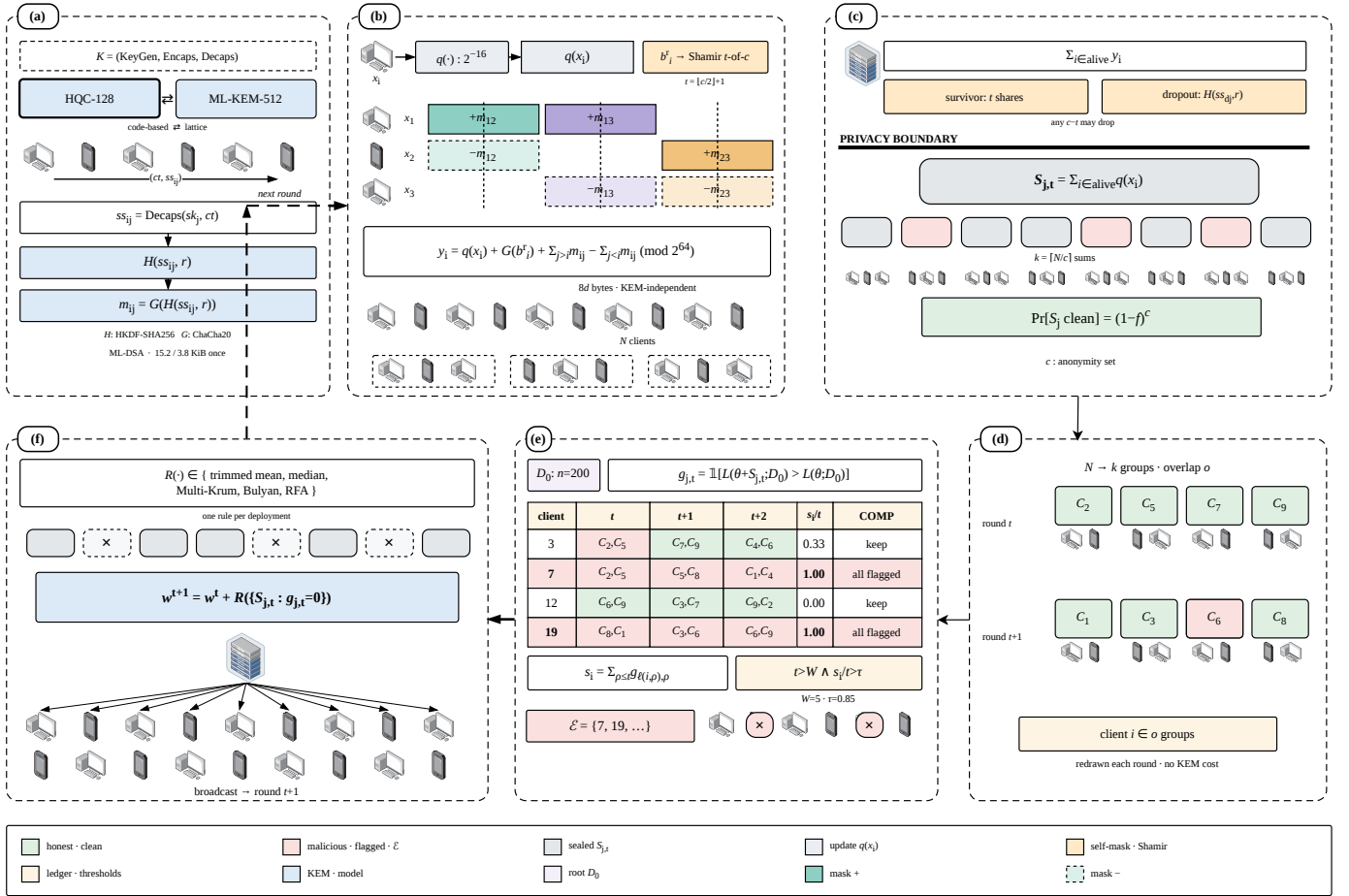


Fig. 1. CB-SAFE end-to-end pipeline: one-time code-based (HQC) key setup, per-round masked cluster aggregation, robust aggregation across cluster sums, and CB-SAFE+ temporal group-testing exclusion.

Models. A 291k-parameter CNN for images and an MLP for Edge-IIoTset. **Attacks.** Sign-flip (with amplification), label-flip, and backdoor, at malicious fractions $f \in \{0.1, 0.2, 0.3\}$. **Baselines.** Undefended mean, trimmed mean, median, Multi-Krum, Bulyan, geometric median (RFA), FLTrust, and the faithful FedGT BCJR decoder. **Environment.** PyTorch 2.5 with liboqs (HQC/ML-KEM) on an NVIDIA RTX 2060; each configuration is repeated over three seeds and reported as mean $\pm$ std. **Training/validation/testing.** One local SGD epoch per round for 50 rounds; a held-out test set measures accuracy each round; a 200-sample server root supports the loss probe.

V. EXPECTED OUTCOMES AND EVALUATION PLAN

Metrics. (i) Test accuracy vs. round and vs. f ; (ii) backdoor attack success rate (ASR, lower is better); (iii) detection quality (attackers identified and honest false positives); and (iv) cryptographic overhead (per-round traffic, one-time key-setup cost, KEM latency). **Expected results.** We expect CB-SAFE+ to track the clean-training baseline under sign-flip while coordinate-wise rules collapse at higher f ; to match FedGT’s attacker identification at a lower false-positive rate while scaling past its $n \leq 31$ limit; and to show that the code-based premium is a *one-time* cost (a few KiB of key

material and a one-time key-setup latency) that is negligible when amortized over training, i.e., post-quantum, code-based confidentiality is affordable in FL. **Validation.** Statistical significance will be assessed by paired t -tests with Holm correction across matched (dataset, f , seed) cells, and cryptographic equivalence by verifying that HQC and ML-KEM yield numerically identical models.

VI. CONCLUSION

The literature reconciles privacy and robustness only in isolation, and post-quantum FL remains lattice-bound. This proposal update targets that gap with CB-SAFE, a KEM-agnostic, code-based post-quantum secure-aggregation framework, and CB-SAFE+, a temporal group-testing defense that identifies hidden attackers without weakening privacy. The detailed methodology and evaluation plan above make the design directly implementable and its claims measurable.

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